Cluster Innovation Centre

Modelling and Network Analysis of Intronic miRNA mediated Gene Expression Regulation

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Abstract

The regulation of gene expression by microRNA has been known for some time. It has also been seen that intronic miRNA govern certain processes in the cell that can generate a feedback mechanism for the amount of proteins that will be formed depending upon the protein requirement by the cell. In this study, we analyse the post-transcriptional regulation by intronic miRNA, present a mathematical model, and establish a network for the interactions.

Mathematical Model

Research has shown that miRNA might bind to mRNA as well as rRNA for the repression of protein production. [1] The presented model leverages the existing models for regulation by intronic miRNA [2], adds to them the component of regulation through binding of intronic miRNA to rRNA. This results in the addition of several new reactions and increases the number of reaction constants on which the regulation depends. The model assumes that DNA is the only source of mRNA as well as rRNA and that the amount of DNA in the cell at all times is known and is a constant.

It is already known that the requirement of proteins by the cell is an important factor in determining the rate of production of mRNA and rRNA, that is, there is a feedback mechanism present in the cell.

Given below is the Overview of the Model (legends on the bottom):

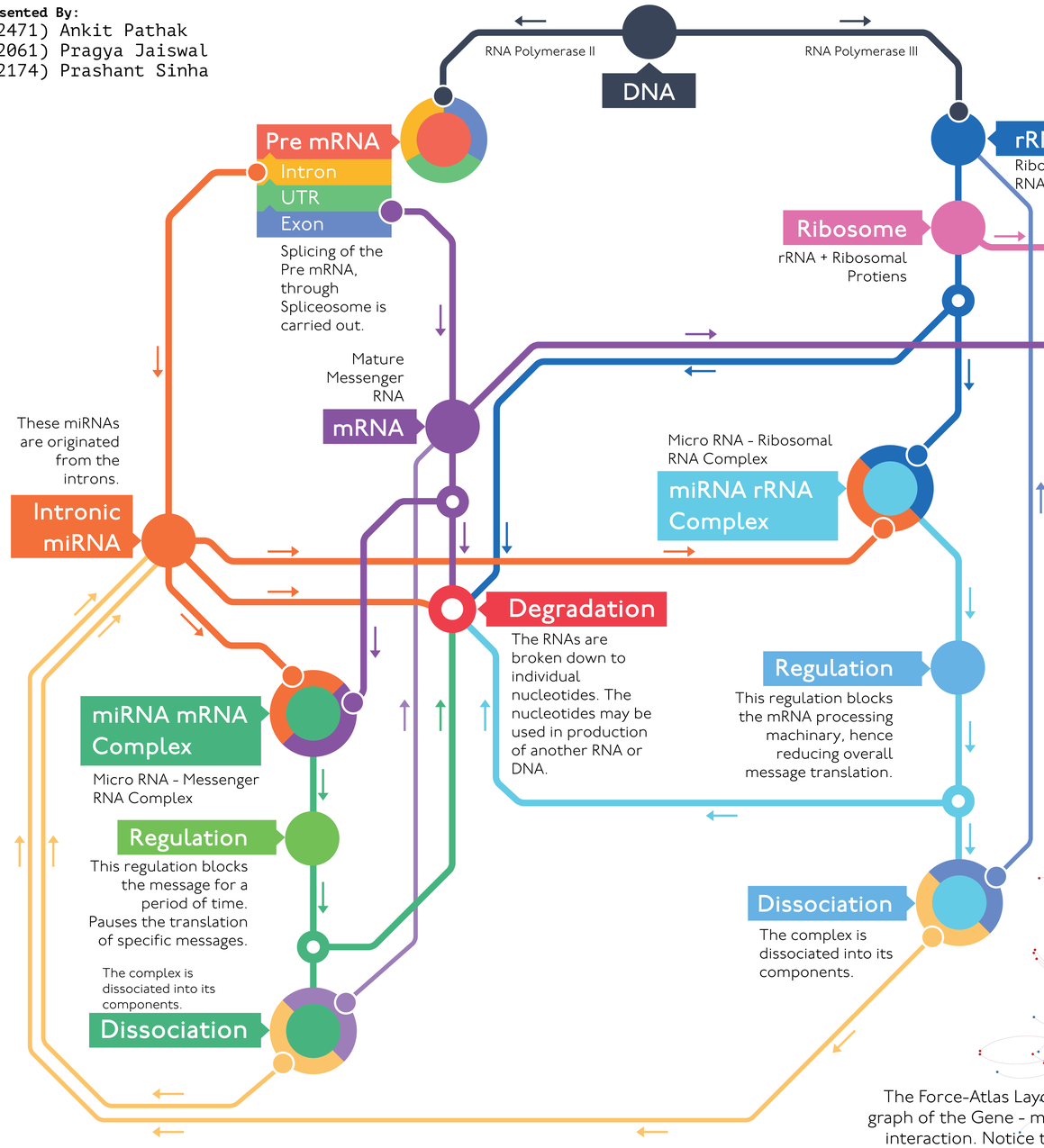
$$\frac{dm[t]}{dt} = k1 + (k12 * m[t]) - (k4 * m[t]) - (k6 * mi[t] * m[t]) - (k8 * m[t])$$

$$\frac{dr[t]}{dt} = k2 + (k3 * r[t]) - (k5 * r[t]) - (k7 * mi[t] * r[t]) - (k9 * r[t])$$

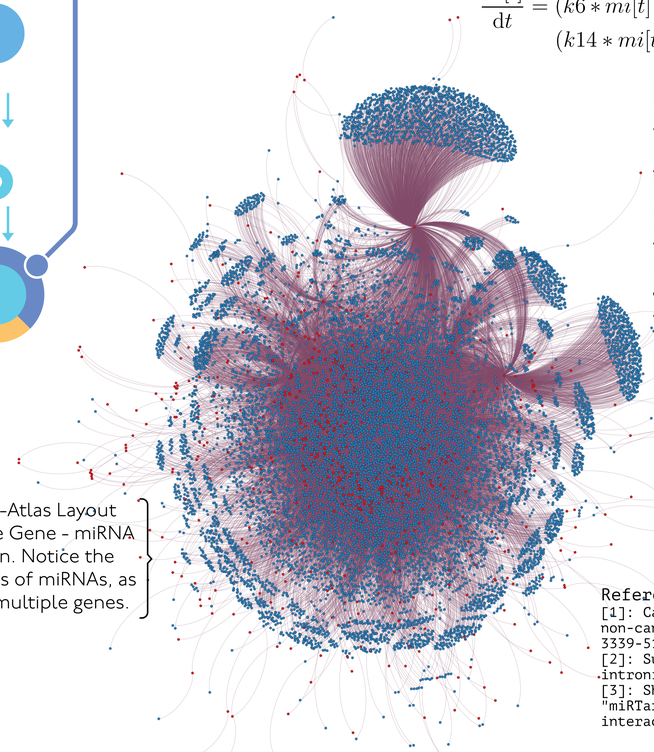
$$\frac{dmi[t]}{dt} = (k3 * m[t]) + (k14 * mi[t]) - (k6 * mi[t] * k7[t]) - (k7 * mi[t] * r[t]) - (k10 * mi[t])$$

$$\frac{dp[t]}{dt} = (k4 * m[t]) + (k5 * r[t]) - (k11 * p[t]) - (k15 * c[t])$$

$$\frac{dc[t]}{dt} = (k6 * mi[t] * m[t]) + (k7 * mi[t] * r[t]) - (k12 * m[t]) - (k3 * r[t]) - (k14 * mi[t])$$



Interaction between complexes and RNA that regulates the production of Protien. Equations on right represent these interactions.



The Force-Atlas Layout graph of the Gene - miRNA interaction. Notice the dense clouds of miRNAs, as regulating multiple genes.

Network Analysis

The attempt to theoretically validate the model and also find the reason to why certain miRNA react with only certain genes led us to the formation of a visualization for the complex biological network between miRNA and the genes they bind to. A collection of 596 miRNAs and 12104 target genes in homo-sapiens [3] was taken.

It is known that few Genes are regulated by a large number of miRNA. The interesting point here is, however, that these Genes are of vital utility for the functioning of cell.

For example, CDKN1A, targetted by 40+ miRNA, is a regulator of cell cycle phase GI.

Mathematical Model Legends

m: mRNA, r: rRNA, mi: miRNA, p: Protein, c: miRNA-mRNA and miRNA-rRNA complexes
k1: mRNA from DNA, k2: rRNA from DNA, k3: miRNA from mRNA, k4: Protein from mRNA processing
k5: Protein from rRNA processinA, k6: miRNA-mRNA complex formatioA, k7: miRNA-rRNA complex formation
k8: Degradation of mRNA, k9: Degradation of rRNA, k10: Degradation of miRNA,
k11: Degradation of Protein, k12: Dissociation of miRNA-mRNA compleA, k13: Dissociation of miRNA-rRNA complex,
k14: Dissociation of miRNA-mRNA and miRNA-rRNA complex, k15: miRNA-mRNA and miRNA-rRNA complex formation

References

[1]: Castellano L., Stebbing J. 2013. "Deep sequencing of small RNAs identifies canonicaland non-canonical miRNA and endogenous siRNAsin mammalian somatic tissues." Nucleic Acids Res 3339-51.
[2]: Sucheta Gokhale, et al. 2012. "A simple method for incorporating dynamic effects of intronic miRNA mediated regulation." Molecular BioSystems 8 (8): 2145-2152.
[3]: Sheng-Da Hsu, Yu-Ting Tseng , Sirjana Shrestha, Yu-Ling Lin, Anas Khaleel et al. 2014. "miRTarBase update 2014:an information source for experimentally validated miRNA-target interactions." Nucleic Acids Research vol 42 (D1) p D78-85.